

Recognizing Header Unit Imports Requires Full Preprocessing

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Abstract

Currently, recognizing header unit imports requires full preprocessing which is problematic for dependency scanning and partial preprocessing. This paper proposes changes that will allow handling such imports with the same degree of preprocessing as `#include` directives.

Revision History

R1 – Add the [Wording](#) section.

Contents

- 1 [Background](#)
- 2 [Proposal](#)
- 3 [Before/After Tables \("Tony Tables"\)](#)
 - 3.1 [Affected Use Cases](#)
 - 3.2 [Unaffected Use Cases](#)

3.3 Unsupported Use Cases

4 Discussion

4.1 Context-Sensitive Keywords

4.2 One Line Requirement

5 Wording

6 Questions and Answers

6.1 Who will be in Cologne to present this paper?

6.2 Is there implementation experience?

6.3 Is there usage experience?

6.4 What shipping vehicle do you target with this proposal?

7 Acknowledgments

1 Background

With the current wording, recognizing a header unit `import` declaration requires performing macros replacement and tokenization of every line in a translation unit. As a representative example, consider the following line:

```
MYDECL import <int>;
```

Whether this is a header unit importation or something else depends on what `MYDECL` expands to. Compare:

```
#define MYDECL int x;
```

```
MYDECL import <int>;
```

And:

```
template <typename> class import;  
#define MYDECL using x =
```

```
MYDECL import <int>;
```

While the second example is contrived, it is valid (again, according to the current wording) because `import` is a context-sensitive keyword.

Requiring such full macro replacement is at a minimum wasteful for header dependency scanning but also may not be something that tools other than compilers can easily and correctly do.

Additionally, several implementations provide support for partial preprocessing (GCC's `-fdirectives-only` and Clang's `-frewrite-includes`) and this requirement is in conflict with the essence of that functionality.

More specifically, GCC is currently unable to support header unit imports in its `-M` (dependency scanning) and `-fdirectives-only` (partial preprocessing) modes because in these modes it does not perform macro replacement in non-directive lines.

While Clang currently performs full preprocessing in its `-M` and `-frewrite-includes` modes, there is agreement that it's not ideal for it to be impossible to correctly extract dependencies without full preprocessing.

Finally, consulting with the developers of `clang-scan-deps` (a Clang-based tool for fast dependency extraction) revealed that this requirement would be problematic for their implementation.

2 Proposal

We propose to further restrict header unit import declarations so that they can be recognized and handled with the same degree of preprocessing as `#include` directives.

Specifically, we propose recognizing a declaration as a header unit import if, additionally to restrictions in [\[cpp.module.1\]](#):

1. It starts with the `import` token or `export import` token sequence that have not been produced by macro replacement.
2. Followed, after macro replacement, by *header-name-tokens*.
3. The entire, single, and only declaration is on one line.

We believe this should not detract much from usability because header imports are replacing `#include` directives where we have the same restrictions.

3 Before/After Tables ("Tony Tables")

3.1 Affected Use Cases

before

```
int x; import <map>; int y;
```

after

```
int x;  
import <map>;  
int y;
```

before

```
import <map>; import <set>;
```

after

```
import <map>;  
import <set>;
```

before

```
export  
import  
<map>;
```

after

```
export import <map>;
```

before

after

```
#ifdef MAYBE_EXPORT
export
#endif
import <map>;
```

```
#ifdef MAYBE_EXPORT
export import <map>;
#else
import <map>;
#endif
```

before

after

```
#define MAYBE_EXPORT export
MAYBE_EXPORT import <map>;
```

```
#define MAYBE_EXPORT
#ifdef MAYBE_EXPORT
export import <map>;
#else
import <map>;
#endif
```

3.2 Unaffected Use Cases

Header unit names are still macro-expanded (similar to `#include`):

```
#define MYMODULE <map>
import MYMODULE;
```

Normal module imports are unaffected:

```
import std.set; using int_set = std::set<int>;
```

3.3 Unsupported Use Cases

With the proposed change the following will no longer be possible:

```
#define MYIMPORT(x) import x
MYIMPORT(<set>);
```

Note also that the following is already impossible (because neither `#include` nor `import`'s closing `;` can be the result of macro replacement):

```
#define IMPORT_OR_INCLUDE(x) ???
IMPORT_OR_INCLUDE(<set>)
```

4 Discussion

4.1 Context-Sensitive Keywords

The proposed change does not fit well with the context-sensitive modules keywords semantics. In the current wording, the context is "wide" taking into account (after macro expansion) previous lines as well as `{}`-nesting. The following examples illustrate the problem:

```
#define MYDECL using x =
MYDECL
import <int>;
```

```
BEGIN_NAMESPACE

template<> class
import<int>;

END_NAMESPACE
```

Our proposed resolution is to adjust context-sensitivity for header unit imports to be based solely on the declaration itself. The fact that `import` should be at the

beginning of the line followed by *header-name-tokens* and terminated with `;` already makes the "pattern" fairly constrained. We could not think of any plausible use-cases for `"` while `<` all seem to boil down to multi-line template-related declarations. And all such cases are easily fixed either by adjusting newlines or with `::`-qualification. For example:

before

```
using x =  
import<int>;  
  
template<> class  
import<int>;
```

after

```
using x =  
::import<int>;  
  
template<>  
class import<int>;
```

Doing a search for `import <` on <https://codesearch.isocpp.org> yielded 2562 matches which unfortunately also included `#import <...>` directives. Doing a search for `#import <` produced 2540 matches. From this we can conclude (though, without seeing the actual code, with low degree of certainty), that there are 20 occurrences of the `import <` token sequence, however, not necessarily at the beginning of the line. We've managed to track at least some of these 20 matches to the Boost.Metaparse library with none of the occurrences being problematic.

4.2 One Line Requirement

Requiring the entire header unit `import` declaration to be on a single line is not strictly necessary. The benefit of this restriction is the simplification of tools that may then be able to reuse the same code to handle both `#include` directives and header unit `import` declarations (at least we found this to be the case for GCC). However, the ability to split the declaration across multiple lines could be beneficial in the presence of attributes. For example (courtesy of Richard Smith):

```
import "foo.h"  
[[clang::import_macros(FOO, BAR, BAZ, QUUX),  
  clang::wrap_in_namespace(foo_namespace)]];
```

5 Wording

Note that according to the direction given at the Cologne meeting, this section extends the original proposal of this paper to all (module and header unit) imports.

In [lex.pptoken]:

preprocessing-token:

header-name

import-keyword

...

(3.3)

Otherwise, the next preprocessing token is the longest sequence of characters that could constitute a preprocessing token, even if that would cause further lexical analysis to fail, except that a *header-name* ([lex.header]) is only formed

(3.3.1)

~~within a *#include* directive ([cpp.include]),~~

after the *include* or *import* preprocessing token in a *#include* ([cpp.include]) or *import* ([cpp.import]) directive, or

(3.3.2)

within a *has-include-expression*, ~~or.~~

~~(3.3.3)~~

~~outside of any preprocessing directive, if applying phase 4 of translation to the sequence of preprocessing tokens produced thus far is valid and results in an *import-seq* ([cpp.module]).~~

Insert new paragraph 4:

4

The *import-keyword* is produced by processing an *import* directive ([cpp.import]) and has no associated grammar productions.

In [basic.link]:

3

A token sequence beginning with `exportopt module` or `exportopt import` and not immediately followed by `::` is never interpreted as the *declaration* of a *top-level-declaration*.

In [cpp]:

1

A preprocessing directive consists of a sequence of preprocessing tokens that satisfies the following constraints: The first token in the sequence, referred to as a directive-introducing token, is a `#` preprocessing token, an `import` preprocessing token, or an `export` preprocessing token immediately followed by an `import` preprocessing token, that (at the start of translation phase 4) either is begins with the first character in the source file (optionally after white space containing no new-line characters) or follows white space containing at least one new-line character. The last token in the sequence is the first new-line character that follows the first token in the sequence.¹⁴⁴ A new-line character ends the preprocessing directive even if it occurs within what would otherwise be an invocation of a function-like macro.

```
...
control-line:
    # include pp-tokens new-line
    exportopt import pp-tokens new-line
    ...
```

4

The only white-space characters that shall appear between preprocessing tokens within a preprocessing directive (from just after the `introducing-#preprocessing` directive-introducing token through just before the terminating new-line character) are space and horizontal-tab (including spaces that have replaced comments or possibly other white-space characters in translation phase 3).

In [cpp.include]:

7

If the header identified by the *header-name* denotes an importable header ([module.import]), the **#include** preprocessing directive is instead replaced by ~~the preprocessing tokens~~ an import directive ([cpp.import]) of the form

```
import header-name ; new-line
```

Rename [cpp.module] "Header units" to [cpp.import] "Header unit importation" and change order so that it appears immediately after [cpp.include] "Source file inclusion".

Move the *pp-balanced-token-seq* and associated productions to [cpp.glob.frag].

In [cpp.import] (previously [cpp.module]):

~~import-seq:~~

~~top-level-token-seq_{opt} export_{opt} import~~

~~top-level-token-seq:~~

~~any pp balanced token seq ending in ; or }~~

pp-import:

~~import header-name pp-import-suffix_{opt} ;~~

~~import header-name-tokens pp-import-suffix_{opt} ;~~

export_{opt} import header-name pp-tokens_{opt} ; new-line

export_{opt} import header-name-tokens pp-tokens_{opt} ; new-line

export_{opt} import pp-tokens ; new-line

~~pp-import-suffix:~~

~~pp-import-suffix-token~~

~~pp-import-suffix pp-import-suffix-token~~

~~pp-import-suffix-token:~~

~~any pp balanced token other than ;~~

The preprocessing tokens after the **import** preprocessing token in the **import control-line** are processed just as in normal text (i.e., each identifier currently defined as a macro name is replaced by its replacement list of preprocessing tokens). A sequence of preprocessing tokens An **import** directive matching the first two forms of a *pp-import* instructs the preprocessor to import macros from the header unit ([module.import]) denoted by the *header-name*. A *pp-import* is only recognized when the sequence of tokens produced by phase 4 of translation up to the **import** token forms an *import-seq*, and the **import** token is not within the *header-name-tokens* or *pp-import-suffix* of another *pp-import*. The **;** preprocessing token terminating a *pp-import* shall not have been produced by macro replacement ([cpp.replace]). The point of macro import for a the first two forms of *pp-import* is immediately after the **;** new-line terminating the *pp-import*. The last form of *pp-import* is only considered if the first two forms did not match.

2

In all three forms of *pp-import* the **import** token is replaced by the *import-keyword* token. Additionally, in the second form of *pp-import*, a *header-name* token is formed as if the *header-name-tokens* were the *pp-tokens* of a **#include** directive. The *header-name-tokens* are replaced by the *header-name* token. [Note: This ensures that imports are treated consistently by the preprocessor and later phases of translation. — end note]

In [module.import]:

module-import-declaration:

```

    exportopt import import-keyword module-name attribute-specifier-
seqopt ;
    exportopt import import-keyword module-partition attribute-
specifier-seqopt ;
    exportopt import import-keyword header-name attribute-specifier-
seqopt ;

```

6 Questions and Answers

6.1 Who will be in Cologne to present this paper?

Boris Kolpackov

6.2 Is there implementation experience?

Yes, an implementation is available in the `boris/c++-modules-ex` GCC branch. This includes working `-fdirectives-only` mode.

One encouraging result of implementing the proposed change was the relative ease of generalizing the `#include` directive handling code in the GCC preprocessor (`libcpp`) and module mapper to also handle header unit imports.

6.3 Is there usage experience?

Yes, the `build2` `build` system implements support for header unit importation relying on this functionality.

6.4 What shipping vehicle do you target with this proposal?

The same as C++ Modules, presumably C++20.

7 Acknowledgments

To our knowledge this issue was first discovered and documented (in the GCC manual) by Nathan Sidwell.

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